

Mini-Mental State Examination: a normative study in Italian elderly population

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The Mini-Mental State Examination (MMSE), a brief test to assess cognitive status, is heavily influenced by age and education. It was administered to 1019 elderly subjects (aged 65–89 years) living in three different Italian cities. A statistical non-linear regression model was built up in order to obtain adjustment coefficients to reduce the influence of demographic variables on the MMSE raw scores. Age and educational level were significantly and independently associated with the MMSE score. Results of a multiple linear regression with transformation of age and education provided adjustment coefficients of the MMSE raw scores. Data from this study will ameliorate the overall reliability of MMSE as a screening test for cognitive impairment in elderly people.

Keywords: Mini-Mental State Examination – Normative study – Elderly subjects – Dementia

INTRODUCTION

The Mini-Mental State Examination (MMSE) (Folstein *et al.*, 1975) is a widely used test adopted to assess the presence of cognitive impairment in elderly people. The reasons for its wide diffusion in clinical and research settings is that the MMSE takes only a few minutes to administer and is reliable enough to evaluate the degree of cognitive impairment and follow the rate of progression of dementing illnesses. Many studies, however, have investigated the limits of this useful tool (Feher *et al.*, 1992; Giordani *et al.*, 1990). One of the problems of using the MMSE with elderly populations is represented by the strong influence on the performance of demographic variables, such as age and education (Jagger *et al.*, 1992). In order to control the confounding effect of age and education, normative data for the MMSE have been published in the last few years (Bleeker *et al.*, 1988; Crum *et al.*, 1993; Heeren *et al.*, 1990; Uhlmann and Larson, 1991). The importance of these studies lies in the possibility of increasing the sensibility and specificity of the MMSE in the detection of cognitive problems (Cummings, 1993). Another aspect that makes these studies important is that the adjusted MMSE score allows a better comparison in cross-sectional surveys of the level of cognitive impairment among subjects of different ages and educational levels.

The aim of this study was to obtain normative data on the MMSE in a large elderly Italian population. Previous studies have already reached the same purpose in non-

Italian subjects. The only Italian normative study on the MMSE (Measso *et al.*, 1993) was based on a population aged 20–79 in which less than 204 out of 906 subjects were aged over 65 years, the age class in which dementia is more common. The size of the sample of elderly subjects considered for this study ($N=1019$) and the fact that they were exclusively over 65 years old and lived at home, represents the basis for this new validation study.

SUBJECTS AND METHODS

Subjects

Subjects included in this study represented the target of three epidemiological surveys conducted in the last years in three different cities of north Italy on the over-70-year-old inhabitants of the cities of Ospitaletto and Coccaglio and over-65s of Tirano. The target population, selected from the Government Registry Office, constituted 1304 subjects. Respondents were 1169 subjects (89.6% of all the inhabitants in the areas investigated). The remaining 10.4% were not interviewed because they refused the interview, at the time of the interview they lived elsewhere, they were in hospital or their location was not correct or available.

Data collection was undertaken between 1992 and 1994 by 30 social workers who underwent a preliminary training course. Demographic variables and MMSE were collected within a multi-dimensional questionnaire. Inter-

TABLE I. Descriptive statistics of performance on MMSE for age intervals and educational levels (mean $\pm$ standard deviation, number of subjects in parentheses)

Educational level						
0–4 years	27.6 $\pm$ 2.0 (6)	27.0 $\pm$ 2.2 (139)	26.2 $\pm$ 2.3 (137)	25.6 $\pm$ 2.5 (100)	24.4 $\pm$ 2.7 (46)	26.2 $\pm$ 2.5 (428)
5–7 years	27.7 $\pm$ 2.3 (48)	27.8 $\pm$ 2.0 (274)	27.3 $\pm$ 2.1 (127)	27.2 $\pm$ 2.2 (53)	26.6 $\pm$ 2.7 (12)	27.6 $\pm$ 2.1 (514)
8–12 years	28.7 $\pm$ 1.1 (14)	28.8 $\pm$ 1.3 (29)	28.8 $\pm$ 1.0 (13)	29.0 $\pm$ 1.2 (6)	27.5 $\pm$ 1.7 (4)	28.7 $\pm$ 1.2 (66)
13–17 years	28.6 $\pm$ 1.1 (3)	28.8 $\pm$ 1.3 (6)	30.0 (1)	27.0 (1)	– (0)	28.7 $\pm$ 1.2 (11)
Total	27.9 $\pm$ 2.0 (71)	27.6 $\pm$ 2.1 (448)	26.8 $\pm$ 2.3 (278)	26.2 $\pm$ 2.5 (160)	25.0 $\pm$ 2.8 (62)	27.0 $\pm$ 2.4 (1019)
Age intervals						
	65–69	70–74	75–79	80–84	85–89	Total

TABLE II. Mini-Mental State Examination score adjustment coefficients for classes of age and education*

Educational level					
0–4 yrs	+0.4	+0.7	+1.0	+1.5	+2.2
5–7 yrs	– 1.1	– 0.7	– 0.3	+0.4	+1.4
8–12 yrs	– 2.0	– 1.6	– 1.0	– 0.3	+0.8
13–17 yrs	– 2.8	– 2.3	– 1.7	– 0.9	+0.3
Age intervals					
	65–69	70–74	75–79	80–84	85–89

Best NLRM for age and education: $2.4 \times \log_{10}$ (93.9—age in years) $\times$ (education in years^{0.29}) + 22.1; adjusted $r^2 = 0.16$.

*Adding the coefficients to the raw MMSE score allows computation of the adjusted MMSE score.

views were conducted door to door. Inter-rater reliability was carried out with three of the social workers for each city and 47 subjects who agreed to be retested at 1-week intervals. Cohen κ for MMSE inter-rater reliability ranged between 0.82 and 0.91. Eighty-eight subjects were excluded from the database because of missing values in one of the variables age, educational level, sex or MMSE score, obtaining a sample of 1081 subjects. A total of 490 subjects (45.4%) were studied in the city of Ospitaletto, 204 (18.8%) in the city of Tirano and 387 (35.8%) in the city of Coccaglio. Subjects scoring less than 21 at the MMSE were excluded from further analysis in order to minimize the number of subjects affected by cognitive problems, obtaining a final sample of 1019 subjects. Table 1 shows the mean MMSE score and the number of subjects for each educational level and each 5-year age interval.

Mini-Mental State Examination—Italian version

An Italian version of the MMSE (Frisoni *et al.*, 1993) was administered to all the subjects included in the study. For

the repetition and the recall of the three words the stimuli “casa” (house), “pane” (bread) and “gatto” (cat) were adopted; in the item *repeat a sentence* we adopted the sentence “sopra la panca la capra campa” (over the bench the goat lives). For backward spelling the word “carne” (meat) was used. Other features of the adopted Italian version were not different from the original one.

Demographic variables

For each subject the age (in years) at the time of the interview, the educational level (in years) and gender (coded as 1 for males and 2 for females) were recorded. Subjects with 1 or 2 years of schooling more than 17 were computed as 17 because they were poorly represented in our sample.

Statistical analysis

Statistical analysis was performed using the SPSS statistical package version 5.0 (SPSS, 1992). A multiple linear regression model (method enter) was applied first in order to assess the independent effects of age, education and sex on the MMSE score. The variables that were found to be associated with the MMSE in this model (*age* and *education*, see Results) were then considered for further analysis.

A non-linear multiple regression (NLMR) model using the formula presented at the foot of Table II contains the age and education best transformations. Adjustment coefficients for the raw MMSE score were calculated on the basis of the NLMR model, subtracting from the predicted values for each education and age class the MMSE mean score of the whole sample. The cut-off dividing the lowest 5% of the population was calculated using 95%-confidence bounds applied to the MMSE adjusted score.

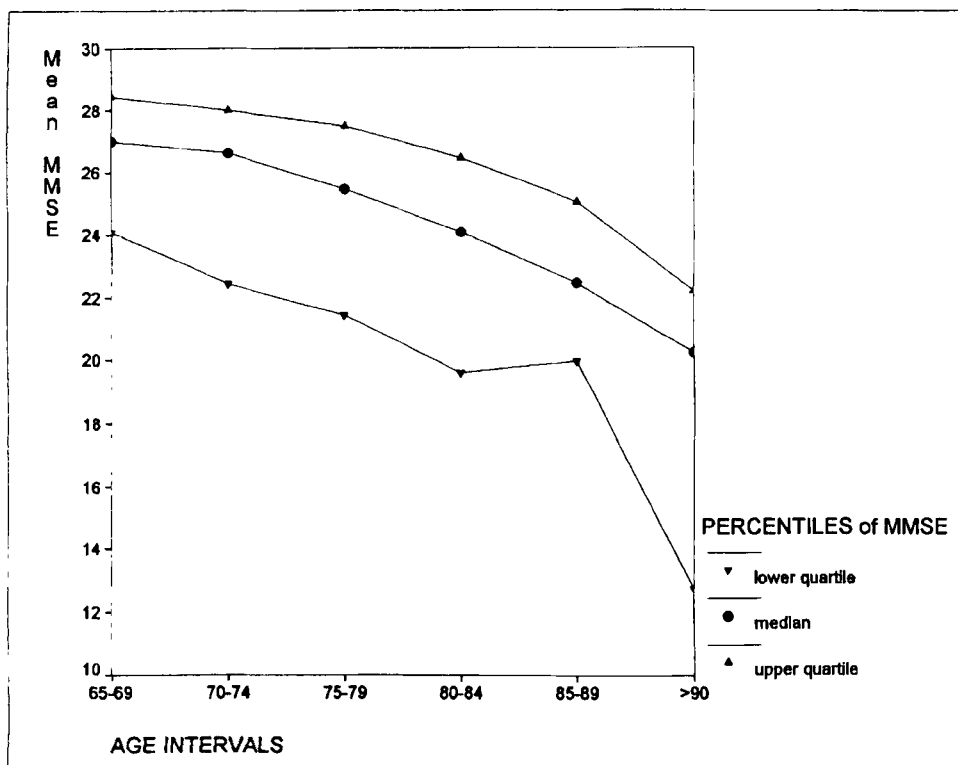


FIG. 1. Distribution across each age interval of the MMSE raw score for median upper and lower quartile in 1019 elderly subjects.

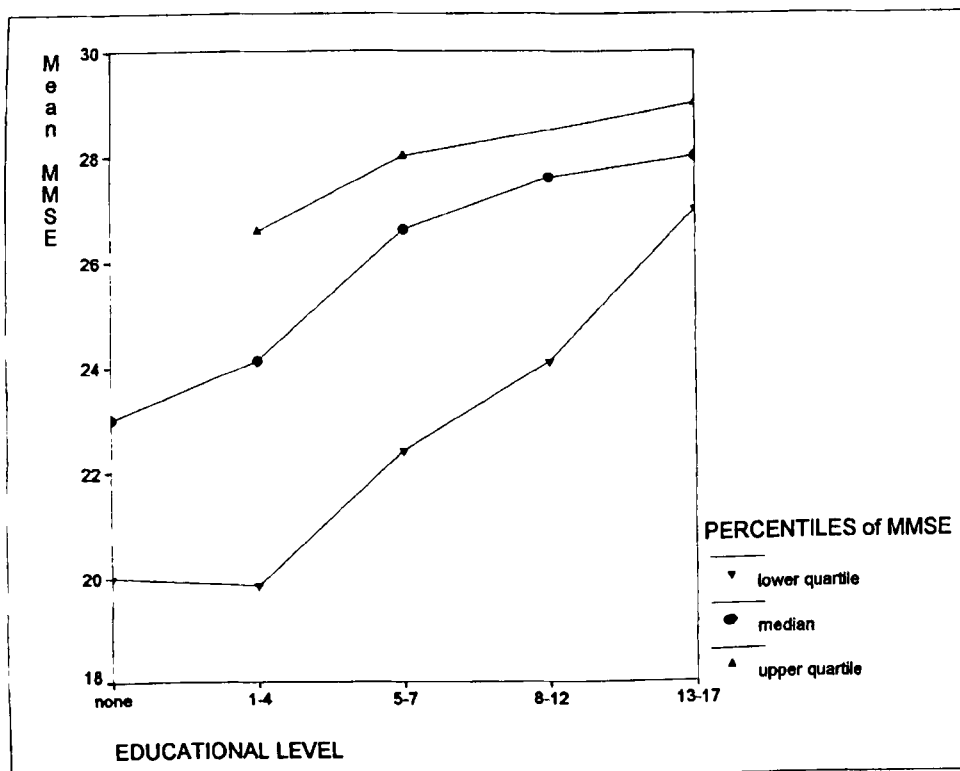


FIG. 2. Distribution across each educational level of MMSE raw score for median upper and lower quartile in 1019 elderly subjects.

RESULTS

The sample selected for this normative data showed a mean age of 75.4 ± 5.4 years and a mean educational level of 5.2 ± 2.5 years. Of the total, 350 subjects (34.3%) were males and 769 were females (65.7%). The mean MMSE score observed in this population was of 27.0 ± 2.4 . A multiple linear association model including sex, age and education was first adopted to assess the relationships between demographic variables and the performance on the MMSE. Results from this analysis showed that age was significantly and inversely associated with MMSE score ($\beta = -0.27$, significance $T < 0.0001$), whereas education was positively associated ($\beta = 0.22$, significance $T < 0.0001$). Sex was not significantly associated with MMSE ($\beta = 0.00$, not significant). A descriptive plot of MMSE distribution across each age interval and educational level is presented for median, upper and lower quartiles of the sample (Figs 1 and 2, respectively). The distribution found in our sample was similar to that of other studies (see Crum *et al.*, 1993). In order to obtain a statistical model able to predict, on the basis of age and education, the MMSE score, these two variables were entered in a NLMR using a logarithmic transformation of age and a quadratic transformation of education as already suggested by other authors (Spinnler and Tognoni, 1987). This model accounted for the 16% of the total variance (adjusted $r^2 = 0.16$). Table 2 shows the adjustment coefficients for the MMSE raw score for each age interval and educational level. The cut-off of 22 represented the value below which 5% of the sample presented the lower adjusted MMSE scores.

DISCUSSION

The MMSE is an important tool for the assessment of cognitive status in elderly subjects who are at higher risk than the young to develop dementing illnesses and for this reason it is widely adopted in neuro-geriatric settings for clinical assessment and research. With the purpose of an Italian validation of the MMSE a wide community-dwelling population was selected using rigorous epidemiological criteria in order to study a sample representative of a non-institutionalized elderly population.

The normative data obtained in this study represent a substantial improvement in the use of the MMSE as a screening test in Italian population since it is based on a wide living-at-home sample which is a common target of screening studies. Furthermore the adjusted MMSE score would be a better measure of global cognitive functioning also in cross sectional studies on demented subjects because the confounding effect of demographic variables is controlled, making comparisons among patients more reliable. Cross-cultural studies would also benefit by the

application of adjustment coefficients reducing cultural differences that are particularly marked between Italy and other Western countries.

A methodological problem, however, was represented by the fact that very old and very educated subjects were not well represented in the sample, reducing the statistical power of NLMR model. Another limitation was the lack of clinical diagnosis of dementia in the subjects examined. According to previous epidemiological studies conducted in Italy, we would expect a prevalence up to 10% of dementia in our population (Rocca *et al.*, 1990). However, since institutionalized subjects as well as those with low MMSE scores were *a priori* excluded by the study, the number of subjects with cognitive problems is expected to be negligible. The NLMR model including the best transformations of age and education is similar to that obtained by Measso *et al.* (1993) on a population including also young subjects. In their study the r^2 derived by the regression was higher than ours. This may be explained by the fact that the smaller range of age of our sample would have reduced the statistical power of the model.

The application of adjustment coefficients derived by the analysis performed in our Italian sample could represent an important strategy to improve the overall reliability of the MMSE not only in research, but also in clinical practice.

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